

Day 1 - Introduction to Cyber Security

Cyber Security Kya Hai?

Bahut log sochte hain:

- Wi-Fi Hacking 
- Instagram Hacking 
- Facebook Password Todna 
- Game Hacking 

 Bas yehi Cyber Security hai...

 Nahi! Bilkul nahi.

Cyber Security isse **bahut bada field** hai.

Real Meaning of Cyber Security

Cyber Security ka matlab hai:

Computers, Mobile Devices, Networks, Applications aur Data ko hackers aur cyber attacks se protect karna.

Simple words me:

 Jaise hum ghar ko chor se bachate hain, waise hi digital duniya me data aur systems ko attackers se bachaya jata hai.

Real Life Example

Hum real life me security use karte hain:

Real Life	Security
Ghar	Lock 
Bank	Guard 
Mobile	Password 
Office	CCTV 

 Same cheez cyber world me bhi hoti hai.

Digital World	Protection
Data	Encryption
Network	Firewall
Account	Password + OTP
Systems	Antivirus

Cyber Security Kya Protect Karta Hai?

1. Data Protection

Sensitive information ko protect karta hai.

Examples:

- Passwords
- Bank details
- Personal photos
- Company files

2. Systems Protection

Computers, Servers aur Networks ko secure rakhta hai.

3. Applications Protection

Apps aur software ko hackers se bachata hai.

Examples:

- Banking Apps
- Instagram
- Amazon
- WhatsApp

4. Financial Assets Protection

Online money aur transactions ko protect karta hai.

Examples:

- UPI
- Credit Cards
- Net Banking

5. People Protection

Users ko scams aur identity theft se bachata hai.

Examples:

- Fake OTP scam
- Phishing emails
- Social media fraud

Cyber Security Important Kyun Hai?

Aaj sab kuch online hai:

- Banking 
- Shopping 
- Government Data 
- Hospitals 
- Companies 

Agar security na ho to:

- Money theft 
- Data leak 
- Identity theft 
- Business shutdown 

ho sakta hai.

 Isliye Cyber Security ek luxury nahi, **necessity** hai.

Common Cyber Attacks

1. Phishing Attack

Fake email/message bhejkar user ko trick karna.

Example:

“Your bank account will be blocked. Click here.”

User link par click karta hai → password chori ho jata hai.

Kaise pehchane?

- Unknown sender
- Urgency create karna

- Fake links
- Spelling mistakes

Example:

- amaz0n.com ❌
- paytm-security.xyz ❌

2. Malware

Malicious/Harmful software.

Types:

- Virus
- Worm
- Trojan
- Spyware

Damage:

- Files delete
- System slow
- Password stealing

3. Ransomware

Attacker data ko encrypt karke lock kar deta hai.

Phir bolta hai:

“Paise do warna data unlock nahi hoga.”

Effects:

- Files inaccessible
- System lock
- Business shutdown

4. Data Breach

Unauthorized person sensitive data access kar leta hai.

Examples:

- User database leak
 - Password leak
 - Credit card leak
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CIA Triad - Foundation of Cyber Security

 Yaha CIA ka matlab spy agency nahi hai.

Cyber Security ka base 3 pillars par bana hai:

C - Confidentiality

Sirf authorized users data dekh sakte hain.

Example:

Sirf tum aur bank tumhara balance dekh sakte ho.

Kaise achieve hota hai?

- Passwords
- Encryption
- Access Control
- 2FA

I - Integrity

Data accurate aur unchanged rehna chahiye.

Example:

Bank balance secretly change nahi hona chahiye.

Kaise achieve hota hai?

- Hashing
- Digital Signatures
- Audit Logs

A - Availability

System jab chahiye tab available hona chahiye.

Example:

Hospital server hamesha online hona chahiye.

Kaise achieve hota hai?

- Backups
- Redundancy
- Anti-DDoS
- Disaster Recovery



Important Rule

Agar CIA Triad ka ek bhi part break ho jaye, to security incident maana jata hai.



Types of Cyber Security Domains

Cyber Security ek huge field hai. Iske andar different domains hote hain.



1. Network Security

Networks ko attacks se protect karta hai.

Includes:

- Firewall
- VPN
- IDS/IPS
- Network Monitoring

Example:

Company Wi-Fi ko hackers se secure karna.



2. Application Security

Apps aur software ko secure banana.

Includes:

- Secure Coding
- Penetration Testing
- Code Review

Example:

Banking app me vulnerabilities find karna.



3. Information Security

Data ko unauthorized access se protect karna.

Focus:

- Confidentiality
- Integrity
- Availability

4. Cloud Security

Cloud platforms ko secure karna.

Examples:

- AWS
- Azure
- Google Cloud

Example:

Cloud storage data ko secure rakhna.

5. Endpoint Security

Individual devices ko secure karna.

Examples:

- Laptop
- Mobile
- Tablet

Tools:

- Antivirus
- EDR

6. IoT Security

Smart devices ko secure karna.

Examples:

- Smart Cameras
- Smart TVs
- Routers
- Sensors

7. SOC (Security Operations Center)

24x7 monitoring team jo attacks detect krta hai

24x7 monitoring team jo attacks detect karti hai.

Work:

- Monitoring
- Alert Handling
- Incident Response



8. Digital Forensics

Cyber crime investigation.

Work:

- Evidence collect karna
- Analyse karna
- Court me present karna



Server vs Node



Server

Powerful system jo services provide karta hai.

Example:

- Website Server
- Database Server



Node

Network ka koi bhi connected device.

Examples:

- Laptop
- Mobile
- Printer



Real Example

Instagram open karte waqt:

- Tumhara mobile = Node
 - Instagram backend = Server
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Authentication vs Authorization

Concept	Meaning
Authentication	Tum kaun ho?
Authorization	Tum kya access kar sakte ho?

Example:

Login karna = Authentication

Admin panel access = Authorization



Encryption

Data ko secret format me convert karna.

Without key data readable nahi hota.

Example:

WhatsApp messages encrypted hote hain.



Firewall

Network ka security guard.

Work:

- Unauthorized traffic block
- Malicious connections stop



VPN

Internet connection ko private aur secure banata hai.

Benefits:

- Privacy
- Secure browsing
- Public Wi-Fi protection



Day 1 Tool - VirusTotal Tool

Online malware scanning tool.

Work:

- Files scan
- URLs scan
- Malware detect

Use:

Suspicious files verify karne ke liye.

Basic Protection Tips

- Strong passwords use karo 
- Same password har jagah mat use karo 
- 2FA enable karo 
- Unknown links avoid karo 
- Antivirus install rakho 
- Software update karo 
- Public Wi-Fi avoid karo  
- Data backup rakho 

Real-Life Banking Example

Component	Risk	Protection
User Account	Hacking	Password + OTP
Server	Cyber Attack	Firewall
Data	Leak	Encryption
User	Phishing	Awareness

Companies Cyber Security Kyun Use Karti Hain?

1. Data Protection

Sensitive company data secure rakhna.

2. Financial Security

Money loss aur fraud avoid karna.

3. Customer Trust

Users ka trust maintain karna.

4. Legal Compliance

Government rules follow karna.

5. Business Continuity

Attacks ke baad bhi business continue rahe.

Hacking vs Ethical Hacking

Type	Meaning
 Hacking	Illegal access aur damage
 Ethical Hacking	Legal security testing

Ethical Hackers:

- Vulnerabilities find karte hain
- Security improve karte hain
- Companies ko protect karte hain

Important Cyber Security Certifications

Ye certifications industry me bahut famous hain.

CHE (Certified Hacker Expert)

Beginner level ethical hacking certification.

Learn:

- Basic Hacking
- Networking
- Security Concepts

eJPT (Junior Penetration Tester)

Beginner-friendly practical pentesting certification.

Focus:

- Web Testing
- Network Pentesting
- Exploitation Basics

OSCP (Offensive Security Certified Professional)

World ki most respected hacking certifications me se ek.

Famous For:

- Practical labs
- Real hacking skills
- Tough exam

Line:

“Try Harder” 🔥

ISO 27001

Information Security Management standard.

Focus:

- Company security policies
- Risk management
- Compliance

GRC (Governance, Risk & Compliance)

Technical hacking se zyada management/security policies side.

Includes:

- Risk Management
- Security Policies
- Compliance

Cyber Security Career Paths

Field	Role
Ethical Hacking	Penetration Tester
SOC	Security Analyst
Cloud Security	Cloud Engineer
Forensics	Cyber Investigator
GRC	Compliance Analyst

Digital Data Life Cycle

Data ka journey:

1. Create 📄
2. Store 💾
3. Use 📊
4. Share 🌐
5. Archive/Delete 🗑️

👉 Har stage par security zaroori hai.

Quick Recap

Topic	Meaning
Cyber Security	Digital protection
CIA Triad	Confidentiality, Integrity, Availability
Phishing	Fake trick attack
Malware	Harmful software
Ransomware	Data lock attack
Firewall	Network security guard
Encryption	Secret data format
VPN	Private secure internet
Ethical Hacking	Legal hacking

Final Thought

Cyber Security sirf hacking nahi hai. Ye digital duniya ko secure rakhne ka system hai.

Golden Rule 🛡️

“Trust mat karo - Pehle verify karo, Phir trust karo.”